

ARCHAI × ACMI

Research concepts for a sovereign, local-first AI collaboration. Each is Research-through-Design: the working prototype, tested in a real gallery, *is* the research, and how we would research, explore and share the data is set out for each.

Seven concepts + further ideas Rob Graham · rob@fineartmedia.tech

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Built and improved with **Huggle** (a transparent multi-agent development log) and secured with **Keytec** (a sovereign secrets & API-key wallet), so implementation never loosens sovereignty.

Seven concepts

Curatorial collection intelligence STAFF-FACING

How does conversational + semantic search change how curators find connections and build exhibitions?

AT ACMI This is the staff-facing twin of the Constellation. ACMI already models the collection as a web of relationships (the lens-driven Constellation, the object-wall adjacencies) for visitors, but curators building exhibitions still discover connections through the CMS, the open collection API, and spreadsheets, largely by keyword and prior knowledge.

RESEARCH Research-through-Design + qualitative: think-aloud sessions, work-diaries and interviews with Collections/Exhibitions staff using ARCHAI's staff mode; log every query and “new connection” moment; compare against baseline catalogue search.

EXPLORE Open-ended semantic and conversational exploration: “show me connections between these,” whole-collection aggregate queries, serendipitous discovery for exhibition development.

SHARE A published “curatorial discovery” design-pattern set; anonymised query logs as an open dataset (with consent); open-source staff tooling; a symposium talk; findings fed back into practice.

Accessibility & multilingual voice VISITOR-FACING

What access gains come from conversational, multilingual, voice-first collection interpretation?

AT ACMI This lands squarely in ACMI's free permanent exhibition, The Story of the Moving Image, where interpretation today is overwhelmingly text-and-screen based, wall labels, object-wall captions, and the Lens's collected-content pages, which structurally excludes low-literacy visitors, blind/low-vision visitors, older visitors, and Naarm/Melbourne's large non-English-first communities (Mandarin, Cantonese, Vietnamese, Arabic, Greek, Italian). ARCHAI adds a conversational, multilingual, voice-first interpretation layer over a defined cluster of ~15-20 works/objects: a visitor speaks or taps in any of a chosen ~8 languages, asks "what am I looking at / why does it matter," and hears a spoken, plain-language answer grounded ONLY in ACMI's own catalogue records and curatorial interpretation.

RESEARCH Mixed-methods visitor studies: comparative trials across modes (conversational, multilingual, voice, plain-language, high-contrast), WCAG + cognitive audits, and observation/interviews with disabled and diverse visitors, co-designed with access consultants.

EXPLORE Visitors explore the same objects across modes; measure comprehension, dwell, independence, satisfaction.

SHARE An open accessibility pattern library + evaluation data; a contribution to the existing accessibility & digital-practice toolkits; open-sourced modes; papers.

Local-LLM evaluation → the sovereign appliance TECHNICAL

Which local models sustain grounded, hallucination-free, fast interpretation on museum-owned hardware?

AT ACMI This is a back-of-house technical experiment, not a visitor demo, its output is the sovereign appliance every later ARCHAI-at-ACMI activity would run on, so it belongs where ACMI already handles technical, controlled, collection-adjacent work: the Blackmagic-supported Media Preservation Lab, run against ACMI's own open collection API. The API is the reason ACMI is the ideal site for this specific question: it exposes clean, structured, field-level records (title, creator, date, credit line, description, Constellation lenses/tags) that give you machine-checkable ground truth, you can score a model's every claim against the actual record field, which you cannot do at Heide (no open API, curated record set only).

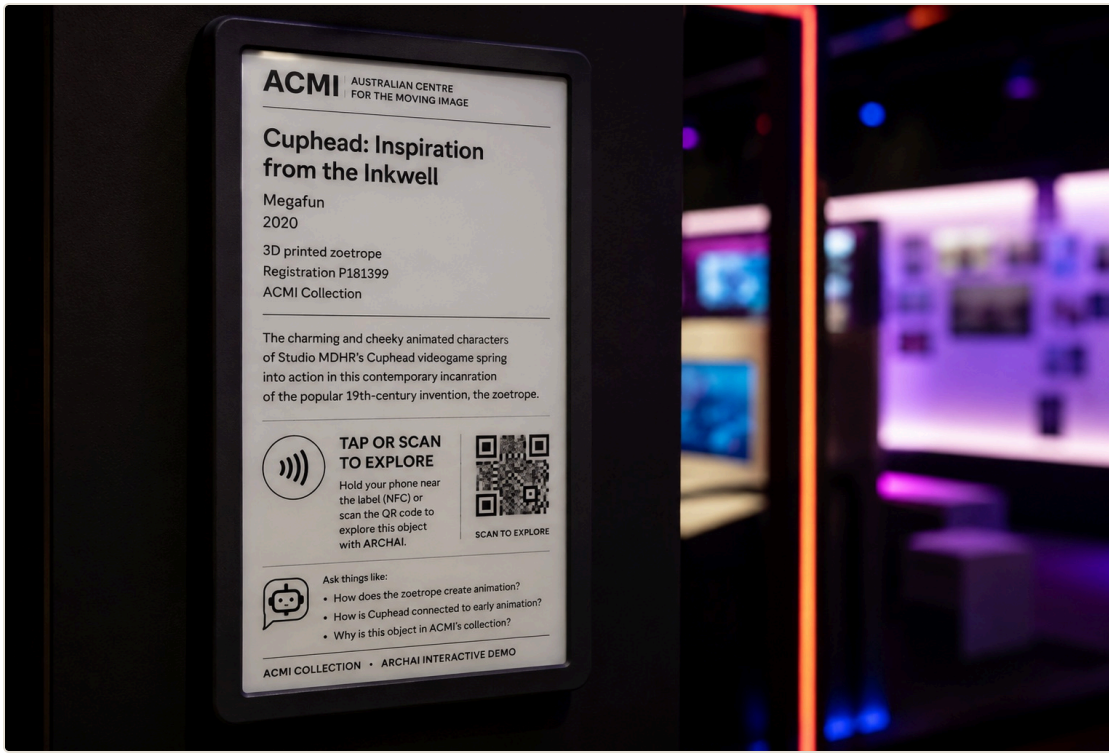
RESEARCH Controlled benchmarking: models × tasks × hardware, scored on grounding fidelity (a five-layer framework, under adversarial prompts), hallucination rate, latency, throughput, power and cost; a reproducible eval harness.

EXPLORE Run the same record set through Qwen / Llama / others; map the fidelity-speed-cost frontier and appliance sizing.

SHARE An open benchmark + reproducible harness; a published hardware reference spec and honest cost model; a sector leaderboard; a technical paper.

E-paper signage + NFC (living labels) PHYSICAL

Can low-power e-paper + NFC turn a static label into living, updatable, conversational interpretation?



Concept mockup, an ARCHAI e-paper label: it redraws from the record, and a tap (NFC) or scan opens a conversation with the object.

AT ACMI Anchor on the Gallery 1 object wall in "The Story of the Moving Image", hundreds of floor-to-ceiling screen-technology objects (cameras, projectors, consoles, phones) whose interpretation is currently thinnest-per-object and locked to a print/production cycle. This is the exact place a "living label" earns its keep.

RESEARCH RtD prototype tested in-gallery via an Audience Lab / spatial-interaction framework: A/B against static labels; observe tap-through, dwell, comprehension; measure power and sustainability.

EXPLORE Visitors explore objects via the phone conversation reached by an NFC tap; staff explore the "label redraws itself from the record" workflow.

SHARE An open hardware/firmware reference + the record → label pipeline; a sustainability/engagement dataset; a replicable "living label" pattern for the sector.

The federated museum “mega brain”

FLAGSHIP

What becomes possible when all open-licensed collections are federated into one local sovereign stack?

AT ACMI This experiment has an unusually exact home at ACMI because ACMI already ships the two things ARCHAI needs and nobody else in the sector has: (1) a genuinely open collection API and CC-licensed metadata on GitHub (maintained by the collection-API developer), and (2) the Constellation, ACMI's own curatorial model of the moving image as a web of connections between works, creators and technologies inside The Story of the Moving Image. The federated sovereign index IS the Constellation computed at sector scale: today the Constellation is hand-authored and single-institution; ARCHAI turns it into a living, multimodal, cross-collection graph by federating ACMI's open records with the other open GLAM screen collections Rob already indexes (Trove/NFSA screen records, Europeana AV, Wikimedia Commons) into one local Qdrant stack.

RESEARCH RtD systems research: build the federated local index across open sources; study cross-collection discovery, provenance handling, and what it surfaces that single-collection search can't; scale + sustainability analysis.

EXPLORE Cross-collection semantic exploration, provenance chains across sources, sector-wide questions.

SHARE An open, federated, provenance-tracked index (respecting every licence and protocol); a contribution to the connecting-archives agenda; an open-sourced federation layer.

The living archive (community provenance)

COMMUNITY PROVENANCE

Can moderated visitor contributions become verified archival metadata without eroding authority?

AT ACMI Anchor it to the Object Wall in the free permanent exhibition "The Story of the Moving Image." The Wall's screen-technology objects (magic lanterns, home-cine and VHS gear, consoles, cameras) are exactly the records with thin, contested or Melbourne-specific provenance, the class of object where a visitor genuinely holds knowledge the institution lacks ("we sold these at Myer," "my father serviced this model," community memory around the videogames and Australian-screen-history holdings). That mismatch is the whole reason to test community provenance here rather than on well-documented film/TV records.

RESEARCH RtD + evaluation: deploy the moderated message wall + curatorial green-star → metadata; measure contribution quality/volume vs curatorial workload; interview curators and contributors.

EXPLORE Visitors explore and add to the object record; curators explore the review queue and verification flow.

SHARE Open-source the community-provenance model + the Saftio safety framework; verified community metadata back into the record, and, with consent, an open contribution dataset.

Cultural protocols as architecture

GOVERNANCE

How can cultural protocols be enforceable properties of the system, not just policy?

AT ACMI The fit is a real, documented gap. ACMI already has the *policy* layer, the First Nations Cultural and Intellectual Property (ICIP) Protocol and the Reconciliation Action Plan, and publicly commits that "First Nations communities have first right of access" to First Nations cultural materials.

RESEARCH Participatory, community-led co-design under CARE + Maïam nayri Wingara; design and stress-test the protocol-enforcing layer (TK Labels, differential access); evaluate whether protocols hold under real use.

EXPLORE Community and curators explore how enforcement behaves (access tiers, labels, exclusions), always community-governed.

SHARE Sharing inverts: the data is not open by default, Authority to Control stays with communities. What's openly shareable is the method, the governance process, and the architecture, co-authored with community.

Further concepts

THROUGH A SCREEN-CULTURE LENS

The Lens as Instrument: Reading the Room, Not the Records

Can a sovereign, local-first model turn The Lens from a souvenir-collection device into a live design instrument, inferring visitor curiosity, dwell, pace, and 'reading style' in the moment, and adapting what the gallery offers next, without profiling individuals or sending a single data point off-site?

The Lens is ACMI's signature interaction and its most under-exploited research asset: today it mostly bookmarks works, but every tap is a latent signal of curiosity and confusion across The Story of the Moving Image. No other museum has this instrument at this scale.

Playable Provenance: A Sovereign Model That Watches Videogames Play Themselves

Can a local multimodal model 'play', watch, and describe ACMI's videogame and time-based works, generating behavioural, procedural, and access documentation from the running artefact itself, so that the collection can be searched, condition-checked, and interpreted by what a work DOES, not just its catalogue metadata?

Videogames are ACMI's crown collection strength and the hardest thing in the building to describe, preserve, and exhibit: a game's meaning lives in its behaviour, and catalogue records can't capture play. This idea treats the executable as the primary source, a model observes emulated/recorded play and produces procedural rhetoric summaries, accessibility notes (difficulty, input demands, photosensitivity risk), and obsolescence/condition signals for the Media Preservation Lab.

The Continuity System: An AV-Aware Model for Keeping the Moving Image Moving

Can a sovereign model fuse ACMI's real-time AV/playback telemetry (Nodel, projection, media servers, e-paper endpoints) with the collection graph to predict exhibition failures, flag format obsolescence before it strands a work, and give the show-floor a plain-language early-warning system, running entirely on local infrastructure?

This is the museum's invisible research problem and Rob's home turf: a screen-culture museum is only as good as its uptime, and the gap between 'this file plays today' and 'this work is preservable' is where born-digital heritage quietly dies. No existing idea touches the operational/AV layer, the mega-brain indexes knowledge, the appliance is the substrate, the e-paper idea is a display surface; none of them reason about playback health, format risk, and show continuity as a live system.

THROUGH A PRESERVATION / BORN-DIGITAL LENS

The Fidelity Witness, does the copy still behave?

Can a local, vision-and-signal model act as an automated conservator's witness, watching an original playback (analogue master, born-digital work, or software artefact) side-by-side with its migrated or emulated surrogate and flagging where the 'significant properties' have drifted: colour and gamma, frame timing and cadence, dropped frames, audio sync, input latency, and characteristic glitch/artefact behaviour?

This attacks the single hardest, least-solved problem in the Media Preservation Lab's own stated workflow, the 'hours of quality assessment on born-digital works' and 'implementing migration schedule and emulation procedures.' Right now fidelity QC is a human squinting at two screens; it doesn't scale to a large collection including thousands of videogames. ACMI is one of very few museums that both emulates software art AND runs a public-facing preservation lab, so a fidelity verifier is directly deployable AND performable behind the glass wall.

Reversible Machines, AI restoration as a logged, undoable conservation treatment

Can generative restoration (denoise, deflicker, dust/scratch removal, frame reconstruction, upscaling) be made ethically defensible for a national collection by treating every AI intervention as a fully documented, reversible conservation treatment, where the model runs locally, each edit is logged with model, weights, parameters and confidence, and the original is never overwritten, turning 'AI cleanup' from a black box into an auditable provenance chain?

Generative restoration is the field's most seductive and most feared frontier: it can rescue decaying Australian screen heritage, but uncontrolled it fabricates detail and destroys authenticity, a scandal risk for a national collection. ACMI is the ideal place to answer it responsibly because it has the conservation rigour (checksums, fixity, migration discipline) to demand reversibility, the born-digital/time-based holdings that actually need it, and a public glass lab where 'showing your working' is part of the mission.

What the Lens Remembers, engagement-informed preservation triage

Can the signals already generated by visitors carrying the Lens, which works they collect, dwell on, replay, and return to, be turned, via an on-premise model, into a preservation prioritisation instrument: telling the Media Preservation Lab which at-risk works the public actually reaches for, and detecting works that fail to load or replay 'in the wild' as an early-warning system for broken emulation and dead media?

The Lens is ACMI's genuinely unique piece of infrastructure, no other museum has a visitor-carried device linking bodies to a moving-image collection at scale, yet its data is used for personalisation, not preservation. Closing that loop is novel: 'preserve what people actually engage with, and know the instant a work stops playing on the floor.' It fuses the two things Rob knows best (his ACMI Visitor Experience / AV / Nodel-integration past and his ARCHAI preservation present) and it's cheaply runnable because the data and the audience-testing apparatus already exist.

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An independent proposal, not affiliated with or endorsed by any institution named. The e-paper label above is a concept mockup.